

Predicting YouTube Video Popularity with Visual, Textual, and Contextual Features Using Multimodal Architectures

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Abstract

We address the problem of predicting YouTube video view counts based on their thumbnail and metadata. Our approach explores multiple multimodal architectures combining visual, textual, and handcrafted features. We evaluate both regression- and classification-based formulations, and propose a hybrid prediction strategy leveraging the strengths of each. In total, we design and implement three distinct architectures tailored to this task. Finally, we show that combining these models leads to improved generalization performance, achieving competitive results on a held-out test set.

1. Introduction

The popularity of YouTube videos is increasingly influenced by how effectively they capture viewers’ attention in a highly saturated platform. Thumbnails, in particular, play a central role as the visual gateway to a video, often determining whether a user clicks or scrolls past. As a result, optimizing thumbnails and associated metadata has become a key concern for content creators and platforms alike.

In this project, we aim to predict the number of views a YouTube video will receive based on its thumbnail and associated metadata. While the image is the primary input, we also leverage optional textual information such as the title and description to improve prediction performance. We frame this as a regression task, using the **Mean Squared Logarithmic Error (MSLE)** as our main loss function to accommodate the highly skewed distribution of view counts.

The dataset used spans videos from 46 curated YouTube channels, published between 2013 and 2023. Our goal is to build models that generalize well to newer content—specifically, videos released in 2024 and 2025—making this work relevant for recommendation systems, content planning, and automated thumbnail evaluation.

Class	View Range
0	0 – 5,000
1	5,000 – 10,000
2	10,000 – 25,000
3	25,000 – 50,000
4	50,000 – 100,000
5	100,000 – 500,000
6	More than 500,000

Table 1. Definition of classification classes based on view count intervals.

2. Problem Formulation

We aim to predict the number of views a YouTube video will receive using only pre-publication data such as the thumbnail, title, and description. This task can be addressed either as a regression problem—predicting the exact view count—or as a classification problem—assigning the video to a predefined popularity range.

In the regression setting, we learn a function f mapping input features x_i to a continuous prediction $\hat{y}_i$ of the true view count y_i . To handle the high variance in the data, we use the **Mean Squared Logarithmic Error (MSLE)**:

$$\text{MSLE}(y, \hat{y}) = \frac{1}{n} \sum_{i=1}^n (\log(1 + y_i) - \log(1 + \hat{y}_i))^2 \quad (1)$$

We also consider a classification approach where view counts are discretized into seven classes, as defined in Table 1. This allows the model to focus on predicting coarse popularity levels, with class-wise statistics optionally used to approximate numerical predictions.

2.1. Dataset Description

The dataset used in this work consists of metadata collected from 46 distinct YouTube channels. It is divided into two subsets:

- A **training set** of 15,482 videos published between 2013 and 2023, each annotated with the corresponding number of views.
- A **test set** of 3,401 videos, published primarily in 2024 and 2025, for which view counts are withheld.

Each video is associated with several metadata fields available prior to publication, including textual information, publication date, and channel identity. A complete description of these attributes is provided in Table 3, included in Appendix 5.

The distribution of videos across years and channels is highly imbalanced, reflecting real-world trends in content frequency and popularity. This imbalance is particularly pronounced both in terms of publication year and view count distribution, as illustrated in Figure 1 and Figure 2, provided in Appendix 5. Such variability makes it challenging to generalize predictions, especially when aiming to extrapolate view performance for unseen videos released in 2024–2025.

2.2. Feature Engineering

Our initial approach relied solely on the video thumbnail and title to predict view counts. However, this limited information proved insufficient to deliver robust performance, particularly for high-variance cases. To address this, we incorporated a broader set of handcrafted features designed to capture additional signals that may influence a video’s popularity.

These features span five main categories: textual attributes of the title (e.g., length, capitalization, punctuation), properties extracted from the description, temporal metadata (such as upload day, hour, or season), channel-level statistics (e.g., average and quantile-based historical views), and sentiment indicators computed from the title. This enriched representation allows the model to exploit structural and contextual patterns beyond raw image and text. A full description of the available handcrafted attributes is provided across Tables 4, 5, 6, and 7, included in Appendix 5.

Our design was inspired by previous work on feature engineering for social media content, particularly the Kaggle project `PogModel: LightGBM + FE + Explanation` by Michal Bogacz [1], which demonstrated the value of structured inputs for view prediction tasks.

3. Our methodology

3.1. Model Architectures

In this section, we present the different architectures developed and tested during the project. Each model reflects a distinct strategy to leverage the available data, including

visual, textual, and handcrafted features. For each, we describe the high-level design and the motivation behind it.

An analysis of their respective strengths, weaknesses, and performance is deferred to Section 4.2.

3.1.1 Multimodal V1: Visual, Textual, and Handcrafted Features

Our first architecture, **Multimodal V1**, combines three modalities to predict video views: the thumbnail image, the video title, and a set of handcrafted features. The thumbnail is processed using a DINOv2 visual backbone, while the title is encoded using a Sentence-BERT (SBERT) model to capture semantic information. In parallel, we compute a vector of handcrafted features capturing textual, temporal, and channel-level statistics, as described in Section 2.2 and detailed in Appendix 5.

The outputs from these three components are concatenated and passed through a multi-layer perceptron (MLP) that produces the final regression output. This architecture allows the model to leverage both learned and structured information, improving prediction accuracy over simpler input combinations.

The structure of Multimodal V1 is illustrated in Figure 3, provided in Appendix 5.

3.1.2 Multimodal V2: Lightweight Architecture

The second architecture, **Multimodal V2**, explores the use of more lightweight and computationally efficient encoders for both image and text inputs. Specifically, we replace the DINOv2 visual encoder with a standard ResNet50 backbone (used in frozen mode), and the Sentence-BERT title encoder with DistilBERT.

The motivation behind these substitutions is twofold: (1) reduce computational cost and inference time, and (2) evaluate whether more compact models can achieve performance close to heavier architectures. ResNet50, while not fine-tuned in our case, remains a strong baseline for image representation, and DistilBERT retains much of the semantic understanding of BERT with significantly fewer parameters.

This design was inspired by an architecture presented by another team during the mid-project checkpoint. Their approach demonstrated promising results with a simpler setup, prompting us to adopt and adapt their idea for further experimentation.

The resulting architecture is illustrated in Figure 5, available in Appendix 5.

3.1.3 Multimodal V3: Residual Attention Fusion

The **Multimodal V3** model extends previous architectures by introducing a more structured and attention-

guided fusion of modalities. In this version, each input stream—thumbnail, title, and handcrafted features—is independently processed through a specialized encoding block before being fused. The goal is to let each modality contribute meaningfully while allowing the network to learn complex cross-modal dependencies.

Each modality is encoded using a lightweight backbone: thumbnails with ResNet50, titles with DistilBERT, and handcrafted features also via a DistilBERT-based encoder. Each encoding branch passes through a **Residual Attention Block (RAB)** followed by a linear projection to 512 dimensions. The resulting three 512-dimensional vectors are then concatenated and passed through a 1D convolutional block and a two-layer MLP that jointly predicts the class and view count.

The motivation behind this architecture is to improve feature interaction across modalities while maintaining a lightweight and tractable model. This design was directly inspired by the approach of Truong Le *et al.* [2], who proposed a residual attention mechanism to enhance multimodal fusion in social media content ranking.

The full architecture is illustrated in Figure 6, and the internal structure of the RAB module is detailed in Figure 7, both provided in Appendix 5.

3.2. Classification and Hybrid Strategy

All three multimodal architectures presented in the previous section were trained to perform both regression and classification simultaneously. This was achieved by modifying the final layer to output, a scalar value for regression and a 7-dimensional logits vector for classification. The latter corresponds to the predefined view count intervals introduced in Table 1.

To enhance granularity, we combined the regression and classification outputs using a **weighted interpolation strategy**. After predicting a class i from the classifier and a continuous estimate x from the regressor, we identify the class j such that $x \in C_j$, where C_j denotes the interval corresponding to class j . Let ℓ_i and ℓ_j be the softmax outputs (probabilities) associated with classes i and j , respectively.

We define the final predicted view count $\hat{y}$ as follows:

- If $i = j$, we directly use the regressor’s output:

$$\hat{y} = x \quad (2)$$

- If $i < j$, we interpolate between the regressed value x and the upper bound of class C_i , denoted $\sup(C_i)$:

$$\hat{y} = x \cdot \ell_j + \sup(C_i) \cdot \ell_i \quad (3)$$

- If $i > j$, we interpolate between x and the lower bound of class C_i , denoted $\inf(C_i)$:

$$\hat{y} = x \cdot \ell_j + \inf(C_i) \cdot \ell_i \quad (4)$$

This rule-based mechanism allows the model to refine continuous predictions with class-level calibration, which we found beneficial in uncertain or boundary cases. A visual representation of this logic is provided in Figure 4.

3.3. Training and Validation

In this section, we describe the training procedure adopted for all models, as well as the validation strategy used to evaluate and compare their performance across different configurations.

3.3.1 Training Procedure

All models were implemented using the PyTorch framework and trained on a single GPU. We used the Adam optimizer with a batch size of 128 and experimented with learning rates of 1×10^{-3} , 1×10^{-4} , and 1×10^{-5} . In some runs, weight decay was also applied to encourage better generalization.

For regression models, we tested several loss functions: the Mean Squared Logarithmic Error (MSLE), which served as our main evaluation metric due to its robustness to scale; the Root Mean Squared Error (RMSE); and the Huber Loss, chosen for its resistance to outliers. For classification-based models, we trained using cross-entropy loss, and additionally evaluated the Focal Loss to address class imbalance.

3.3.2 Validation Strategy

Our initial validation set consisted of 1,500 videos sampled from the years 2018 to 2023. The sampling followed the temporal distribution illustrated in Appendix 5, with the goal of constructing a validation set that closely reflects the time span of the test set (composed of videos from 2024–2025). This strategy was intended to better simulate deployment conditions and assess generalization to more recent content. This validation protocol was applied consistently for the first two architectures: Multimodal V1 and V2.

For the Multimodal V3 model, we experimented with an alternative validation set specifically designed to align with the classification formulation. It included exactly 100 videos from each of the 7 defined view count classes, all drawn from videos published in 2023. This class-balanced split was intended to provide a more relevant evaluation of the classifier’s ability to distinguish between popularity ranges.

However, due to an unnoticed implementation error, the Multimodal V3 model was inadvertently trained on the entire training dataset, including the samples originally reserved for validation. As a result, we were unable to properly assess the effectiveness of the class-balanced validation

Model	Trainable Parameters
Multimodal V1	1.03M
Multimodal V2	3.59M
Multimodal V3	16.4M

Table 2. Number of trainable parameters for each model architecture.

split in that case, which limits the interpretability of Multimodal V3’s reported results.

3.3.3 Implementation Details

All experiments were conducted using the PyTorch framework, which served as the primary platform for model development and training. Models were trained on virtual machines provided by the institution, specifically those equipped with NVIDIA RTX A5000 GPUs.

The number of trainable parameters varied significantly across architectures, reflecting the differing complexity of their components. A summary of model sizes is provided in Table 2.

4. Experiments and Results

4.1. Experimental Setup

All models were trained and validated following the procedure described in Section 3.3. We evaluated all experiments using the MSLE loss. Models were trained for a different number of epochs and with varying learning rates and weight decay values, in order to obtain the best possible performance for each architecture.

4.2. Results

We evaluated all models on the test set using the Mean Squared Logarithmic Error (MSLE).

Multimodal V1. Incorporating handcrafted features led to a notable gain in performance. The Multimodal V1 model achieved an MSLE of 3.47, demonstrating the importance of structured information — such as temporal patterns, channel-level statistics, and sentiment — in view prediction tasks. These features allowed the model to capture higher-level trends that are difficult to extract from raw image and text inputs alone.

However, we observed two significant limitations. First, the model consistently underpredicted videos with very high view counts (in the hundreds of thousands or millions), which indicates a poor fit for long-tailed distributions. Second, when trained using alternative loss functions such as RMSE and Huber loss, the model exhibited significant instability, with large fluctuations in the training and validation curves.

To mitigate these fluctuations, we attempted to rescale the target variable by dividing all view counts by 1,000 before training. While this normalization reduced the numerical scale of the targets, it did not improve model performance. A plausible explanation is that after scaling, videos with modest view counts (e.g., a few thousand views) were effectively treated as close to zero. This loss of resolution in the lower range likely made it more difficult for the model to distinguish between low- and mid-range performances, leading to poorer discrimination among common examples.

Towards Hybrid Modeling. These limitations motivated a shift toward classification-based modeling. Initially, we attempted a binary classification setup that distinguishes between high- and low-view videos, using a threshold of 100,000 views. The idea was to train two separate regressors tailored to each group. However, this approach was hindered by the low accuracy of the classifier, which never exceeded 46% accuracy. Consequently, the separation was too noisy to serve as a reliable basis for downstream regression.

We also experimented with conventional machine learning classifiers, including Random Forests, Logistic Regression, XGBoost, and Support Vector Machines. Despite careful hyperparameter tuning, none of these models achieved better accuracy than the deep learning-based classifiers. Their performance failed to capture the complex interactions between features. As a result, this direction was ultimately abandoned.

We therefore adopted the hybrid classification-regression strategy described in Section 3.2, which interpolates between class-level estimates and continuous predictions. This approach allows the model to leverage both categorical and continuous supervision, while mitigating the weaknesses of each. As we show below, it contributed to performance gains across all subsequent architectures.

Multimodal V2. This version aimed to reduce computational cost by replacing DINOv2 with ResNet50. Despite the model’s simpler structure, it reached an MSLE of 3.41 — slightly better than V1. However, the most notable advantage was the fourfold reduction in training time. This made Multimodal V2 a compelling trade-off between efficiency and accuracy, particularly for resource-constrained applications or large-scale training scenarios.

Multimodal V3. The final model, based on residual attention fusion, achieved the best performance with an MSLE of 3.25. This architecture demonstrated superior capacity to integrate heterogeneous modalities by encoding each separately before fusing them with attention-guided blocks. The improved alignment and interaction across thumbnail, title, and handcrafted features allowed the model to better capture

non-linear dependencies. In particular, the fusion scheme helped resolve ambiguous or conflicting signals, such as when a high-quality thumbnail accompanies a generic title.

We also experimented with two additional variations inspired by the approach of Truong Le *et al.* [2]. First, we replaced the ResNet50 visual encoder with EfficientNet, but observed a drop in performance, suggesting that the additional capacity did not translate into better generalization in our setting. Second, we introduced a fourth modality corresponding to tags inferred from the title and description, as suggested in their work. However, this extension did not yield significant gains and was eventually discarded.

Overall, these results highlight the progressive improvements achieved through model refinement, feature enrichment, and better fusion strategies. They also underscore the trade-offs between complexity, interpretability, and computational efficiency across the different model versions.

4.3. Final Submission

Our final submission in the project consists of an ensemble of the most promising models developed throughout our experiments. Specifically, it combines the predictions of multiple regression models through simple averaging, as well as the class probabilities (logits) from the classification heads of the same models. This dual fusion strategy was designed to leverage the individual strengths of each architecture and to reduce variance in the final output.

For the regression component, we averaged the predicted view counts from the best-performing models, each trained under its optimal configuration. For the classification component, we averaged the softmax outputs across the corresponding classifiers before applying the hybrid interpolation strategy described in Section 3.2.

The goal of this ensemble approach is to exploit the complementarity between models — for instance, one model may be more robust on low-range views, while another performs better on high-view outliers. By aggregating their outputs, we aimed to build a more reliable and generalizable predictor.

This final submission achieved a significant performance gain, with an MSLE of 2.80 on the test set — the best result obtained across all our experiments.

5. Conclusion and Future Work

Our analysis reveals several key findings that shed light on the complexities of predicting YouTube video popularity.

We found that models relying solely on basic visual and textual inputs—thumbnails and titles—struggled to capture the nuanced factors that drive viewership. These simple modalities proved insufficient for encoding the rich contextual information needed for accurate predictions. However, when we enriched our models with carefully engineered

features from temporal metadata, descriptive features extracted from the title and description, and channel statistics, we saw marked improvements in performance. This enhancement was particularly pronounced in the middle range of the view distribution, where prediction signals are more subtle and harder to detect.

Interestingly, our work with simpler architectures like Multimodal V2 showed that computational efficiency need not come at the expense of accuracy. These lightweight models performed surprisingly well, suggesting that architectural complexity alone does not guarantee better results. We also discovered that combining classification and regression approaches through a hybrid strategy proved particularly effective for handling the long-tailed nature of view distributions, where traditional single-method approaches often struggle with stability and bias.

While these results are encouraging, several areas warrant further investigation. The classification component of our hybrid framework represents a clear opportunity for improvement, as its accuracy directly impacts the performance of the specialized regressors we train for different popularity ranges. Another important direction is the fine-tuning of hyperparameters, which in our experiments was not conducted exhaustively. A more systematic exploration of learning rates, weight decay, batch sizes, and model-specific settings could reveal further performance gains.

We also see potential in treating view patterns as time series data, which could capture important temporal dynamics, especially when analyzed at the channel level. Finally, expanding our approach to handle multilingual content through language-agnostic embeddings would significantly broaden its applicability, particularly in regions where non-English content dominates the platform.

References

- [1] Michal Bogacz. Pogmodel: Lightgbm + fe + explanation. <https://www.kaggle.com/code/michau96/pogmodel-lightgbm-fe-explanation>, 2022. Kaggle Forum Contribution. 2
- [2] Truong Le et al. Entube: Exploring key video features for advancing youtube engagement. 2025. 3, 5

Appendix

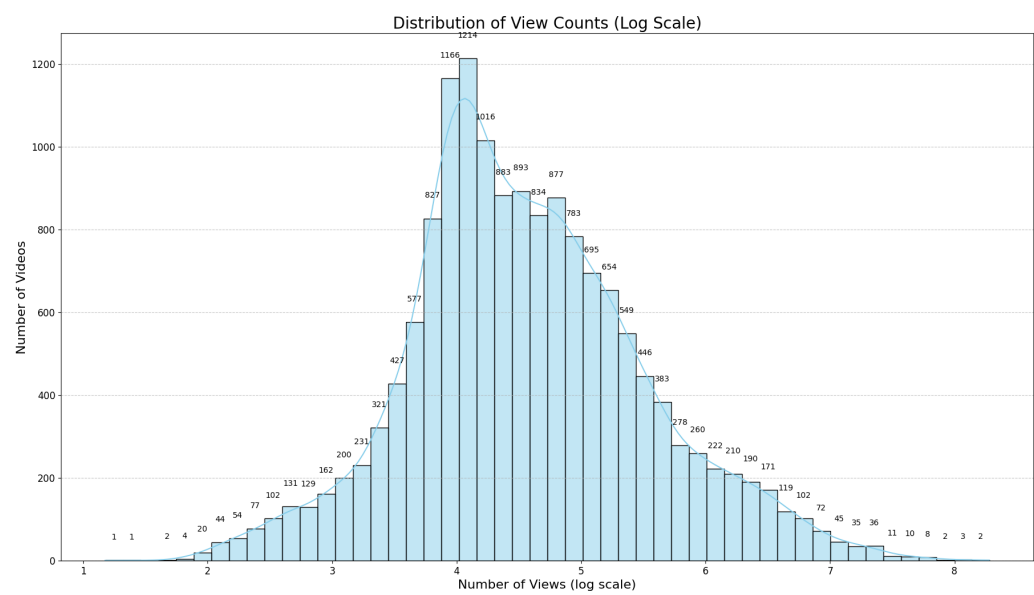


Figure 1. Distribution of view counts (logarithmic scale) across all videos in the dataset.

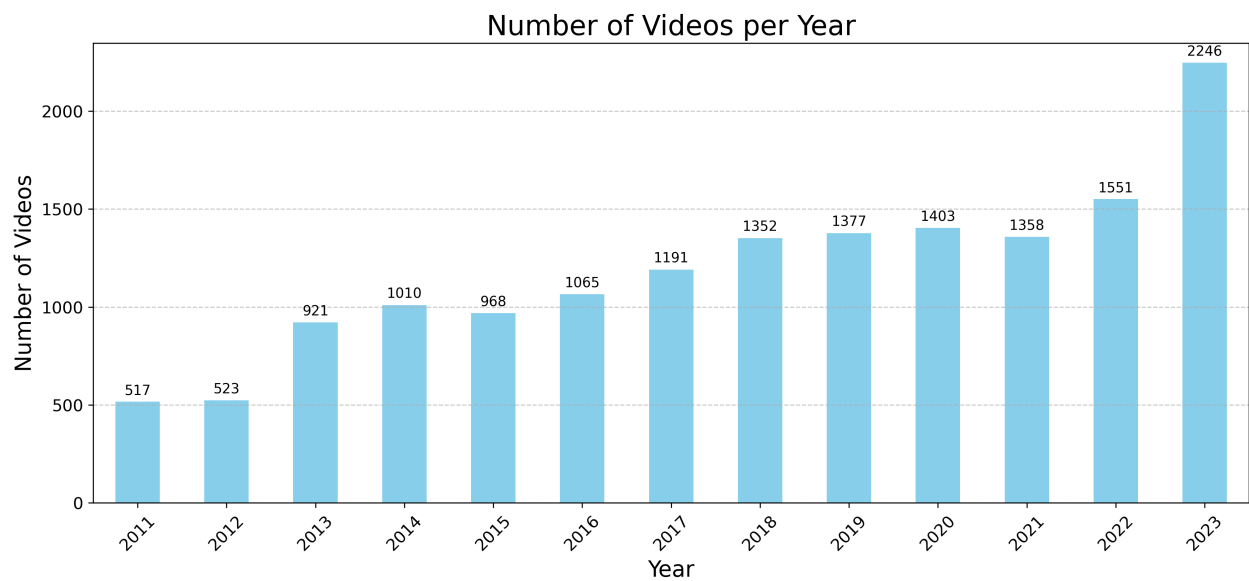


Figure 2. Number of available videos per year (2011–2023).

Attribute	Description
id	Unique identifier of the video
title	Title of the video
description	Description text of the video
channel	Name of the YouTube channel
date	Full publication date
year	Year extracted from the publication date
views	View count (only available in training set)

Table 3. Description of dataset attributes used as input features.

Feature Name	Description
title char count	Number of characters in the title
title word count	Number of words in the title
title unique ratio	Ratio of unique words
title upper ratio	Uppercase character ratio
title has question	Presence of question mark
title has exclamation	Presence of exclamation mark
title hashtag count	Number of hashtags
title avg word length	Average word length
title stopword ratio	Proportion of stopwords
title has number	Contains a numerical digit
title starts with number	Starts with a number
title has capslock word	Contains all-caps word
title emojis count	Number of emojis
title special char count	Number of special characters
title word overlap desc	Overlap with description

Table 4. Title-based handcrafted features.

Feature Name	Description
description char count	Number of characters
description word count	Number of words
description hashtag count	Number of hashtags
description link count	Number of links
description mention count	Number of mentions
description emojis count	Number of emojis
description link ratio	Ratio of links to words
description sentiment polarity	Sentiment polarity score

Table 5. Description-based handcrafted features.

Feature Name	Description
publish year	Year of publication
publish month	Month of publication
publish dayofweek	Day of the week
publish hour	Hour of publication
publish weekofyear	Week number of the year
video age days	Days since upload
is weekend publish	Published on a weekend
publish part of day	Morning, afternoon, etc.
is holiday season	Holiday period indicator

Table 6. Temporal handcrafted features.

Feature Name	Description
channel video count	Number of videos published
channel mean views	Mean view count
channel median views	Median view count
channel q25 views	First quartile view count
channel q75 views	Third quartile view count
channel min views	Minimum view count
channel max views	Maximum view count
channel iqr views	Interquartile range
title polarity	Sentiment polarity of the title
title subjectivity	Subjectivity score of the title

Table 7. Channel-level and sentiment features.

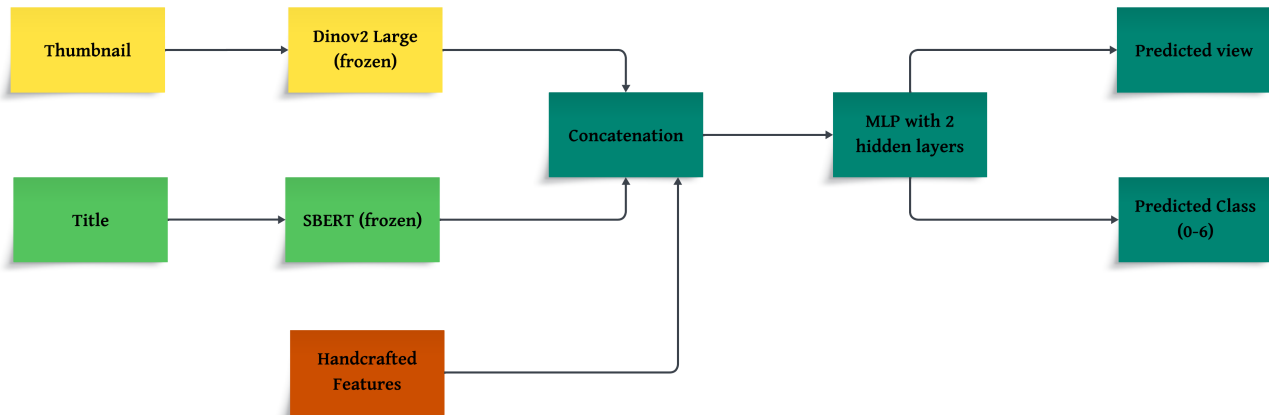


Figure 3. Architecture of Multimodal V1: handcrafted features are concatenated with the image and title embeddings before prediction.

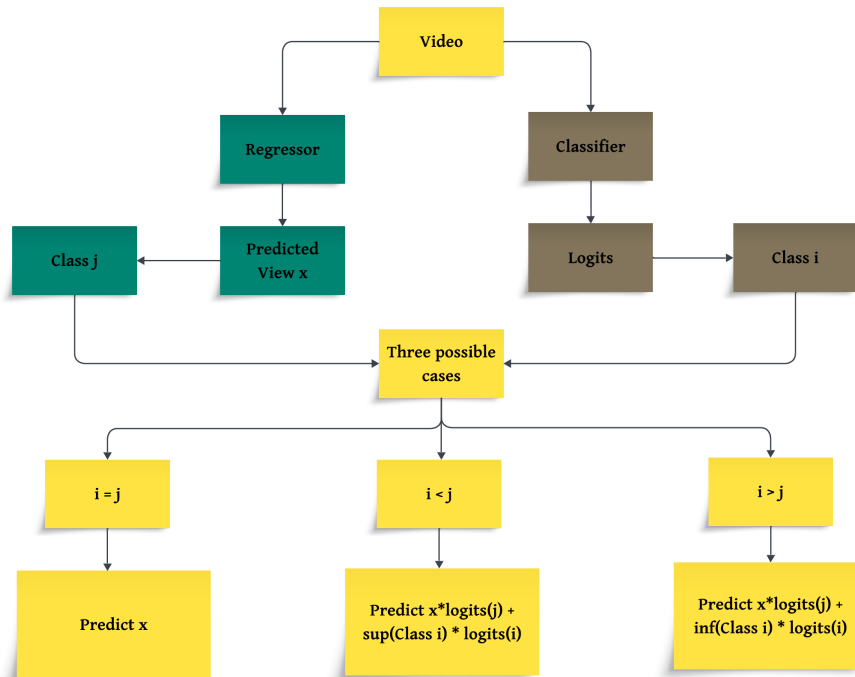


Figure 4. Decision flow for combining regression and classification outputs. The model uses logic based on predicted class and continuous output to adjust the final view estimate.

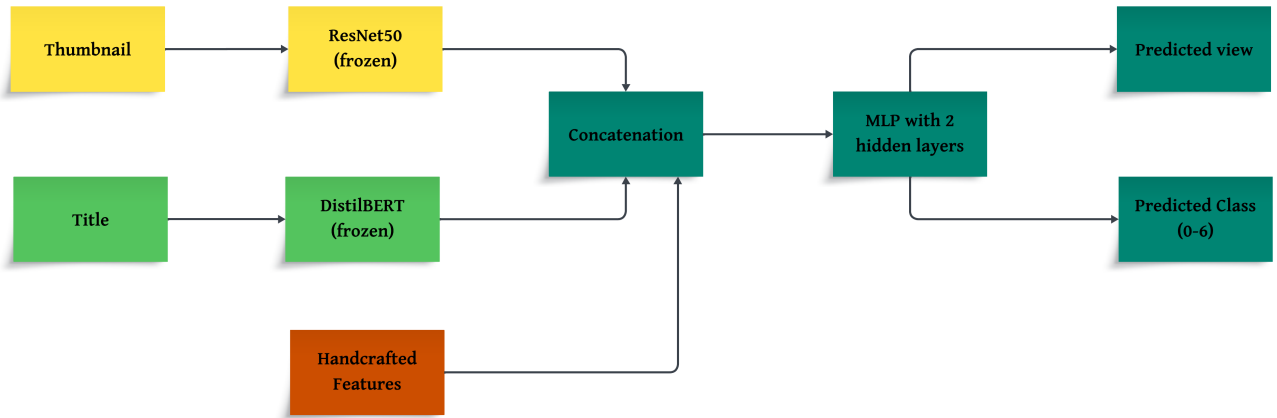


Figure 5. Architecture of Multimodal V2: ResNet50 for thumbnail encoding and DistilBERT for title embedding.

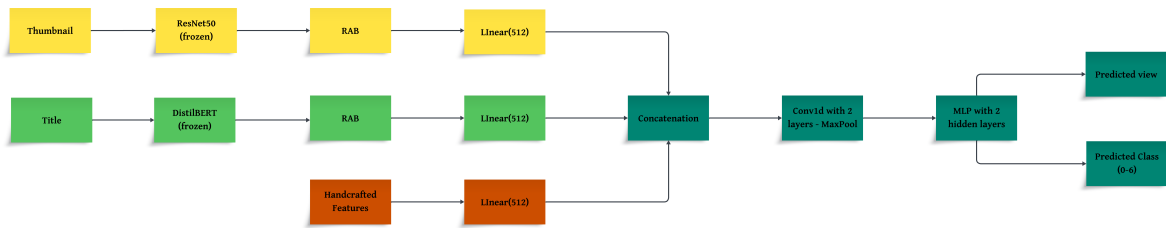


Figure 6. Architecture of Multimodal V3 with residual attention fusion. Each modality is encoded separately and then fused via 1D convolutions before final prediction.

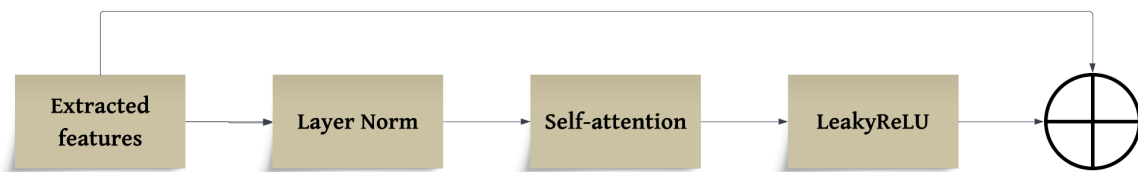


Figure 7. Architecture of the Residual Attention Block (RAB) used in Multimodal V3.

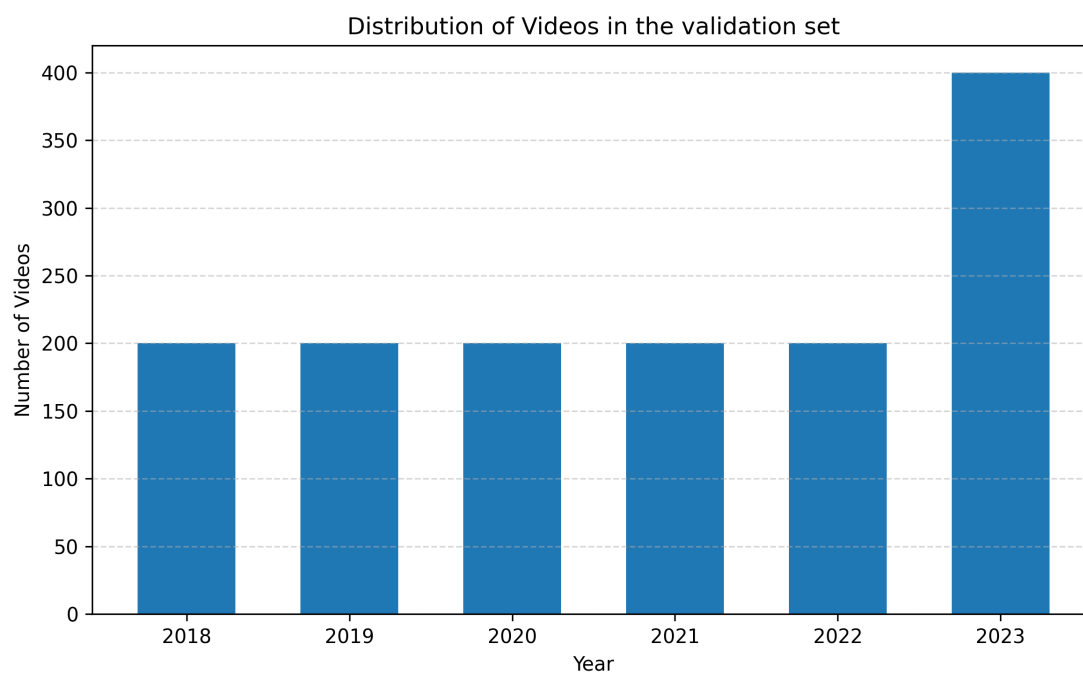


Figure 8. Year-wise distribution of the initial validation set (2018–2023), used for Multimodal V1 and V2.